Splitting the Matroid Determinant

Clara Briand Leonie Kayser Julian Weigert

October 24, 2026

The principal matroid determinant E_L of a linear space $L \subseteq \mathbb{P}^n$ has been introduced in recent work by Matsubara-Heo and Telen. In this paper, we give the complete factorization of this polynomial into its irreducible components, proving a conjecture of the aforementioned authors. Our methods rely on bounding local multiplicities and an étale-local description of the strata of reciprocal linear spaces developed by Elias, Proudfoot and Wakefield. Collecting degrees in the factorization of E_L yields a geometric interpretation of a relation among matroid invariants related to Möbius inversion. We also discuss analogous questions for other coordinate-wise powers of linear spaces.

Contents

1 Introduction **1**

2 Preliminaries **3**
2.1 Generalized matroid determinants 3
2.2 Local multiplicity 8

3 Proof of the main theorem **11**
3.1 A lower bound 12
3.2 Local multiplicities of L^{-k} 15
3.3 Matching the lower bound 19

4 Other powers of linear spaces **20**

References **22**

1 Introduction

The theory of discriminants, resultants and determinants, carefully developed by Gelfand, Kapranov and Zelevinsky in [GKZ94], plays a central role in algebraic geometry, particularly in applications. Loosely speaking, a discriminant is a polynomial that vanishes

whenever a geometric object of study behaves non-generically. A particularly rich class of examples arises from toric geometry: given a finite collection of lattice points $A = \{a_0, \dots, a_n\} \in \mathbb{Z}^d$, the equation Δ_A of the projective dual $X_A^\vee$ of the associated toric variety vanishes on those points $(c_a)_{a\in A} \in (\mathbb{P}^n)^*$ for which the hypersurface defined by the polynomial $f_c(x) \coloneqq \sum_{a\in A} c_a x^a \in \mathbb{C}[x_0^{\pm 1}, \dots, x_n^{\pm 1}]$ is singular.

A beautiful aspect of the toric case is the behavior of the associated *A-discriminants* and *A-determinants* under the natural stratification of toric varieties into torus orbits [GKZ94, Chapter 10]. A similarly combinatorial stratification arises when studying *reciprocal linear spaces* L^{-1}, the image closure of linear spaces under the Cremona transform. Such reciprocal linear spaces decompose similarly according to the flats of the associated matroid $M \coloneqq \mathrm{M}(L)$ [PS06]. Building on this idea, Matsubara-Heo and Telen propose the study of the *principal matroid determinant* $E_L \subseteq (\mathbb{P}^n)^*$ [MT26, Definition 3.2], the central object of study in this article. This hypersurface is a substitution of the Chow form of the projective variety L^{-2}. A hyperplane $H \in (\mathbb{P}^n)^*$ belongs to the vanishing locus of E_L whenever the intersection $(\mathbb{C}^\times)^n \cap L^{-1} \cap H$ does not have the generic Euler characteristic [MT26, Theorem 1.2(4)]. This directly parallels similar result in the toric setup due to Esterov [Est13, Theorem 1.8]. The construction of the principal matroid determinant naturally generalizes to negative powers of linear spaces. We thus define *generalized matroid determinants* $E_{L,-k}$ for any $k \geq 2$ (see Definition 2.7 below for details), with $E_{L,-2} = E_L$ being the principal matroid determinant.

While many properties of principal A-determinants directly generalize to principal matroid determinants, one particularly interesting property remained open in [MT26]: the factorization of the principal matroid determinant $E_{L,-k}$ as a product of matroid discriminants $\Delta(L_F^{-k+1})$ with multiplicities. The aim of this article is to fill this gap by proving the following theorem.

Theorem A. *Let $L \subseteq \mathbb{P}^n$ be a linear space with loopless matroid $M = \mathrm{M}(L)$ and let $k \geq 2$. The generalized matroid determinant $E_{L,-k}$ decomposes into irreducible factors as*

$$E_{L,-k} = \prod_{F \text{ conn. flat of } M} \Delta(L_F^{-k+1})^{m_F}, \qquad m_F = k^{\mathrm{rk}(M/F)-c(M)+1}\mu^+(M/F).$$

The multiplicity m_F further equals the local multiplicity of L^{-k} along the stratum indexed by F. In particular, [MT26, Conjecture 7.1] holds true.

While the components of the hypersurface $\mathrm{V}(E_{L,-k})$ were already known set-theoretically for $k = 2$ [MT26, Theorem 1.2], our work contributes a precise description of the exponents m_F, both in terms of matroid invariants and in terms of the local geometry of L^{-k} for any $k \geq 2$. Our main proof can be outlined as follows.

(1) In Section 3.1 we show that m_F is a lower bound for the multiplicity of the factor $\Delta(L_F^{-k+1})$ inside $E_{L,-k}$. To obtain this lower bound we use the definition of $E_{L,-k}$ as a pullback of the Chow form of L^{-k}. We first track how local multiplicities may change when passing from L^{-k} to the Chow incidence variety and then through the projection to the Chow form and its pullback to $E_{L,-k}$. In this way, we find that the local multiplicity of L^{-k} along the stratum indexed by a flat F is a lower bound to the multiplicity of the factor $\Delta(L_F^{-k+1})$ in $E_{L,-k}$.

(2) Next, in Section 3.2 we compute the local multiplicity of L^{-k} along every flat stratum exactly: We first translate this into a local computation on L^{-1} and then use that locally around a point in a flat stratum, L^{-1} splits into a product of smaller reciprocal linear spaces via an étale map due to [EPW16].

(3) Finally, in Section 3.3 we argue that the multiplicities predicted by Theorem A sum up to the correct degree and hence the lower bounds must all be equalities. For the last step we use Möbius inversion, as well as the formulas for the degrees of $\Delta(L_F^{-k+1})$ proven in our previous work [BKW26, Corollary 1.4].

This manuscript is organized around the proof of the main result in Section 3. The preliminary Section 2 recalls and generalizes the definition of principal matroid determinants, comparing it to the toric story, as well as providing a primer on local multiplicities in commutative algebra. After the proof of the main theorem, in Section 4 we make first steps towards studying the analogously constructed generalized matroid determinants of L^k, $k \geq -1$ and state analogous conjectures about their factorization.

2 Preliminaries

Throughout this article, we work over an algebraically closed field $\mathbb{K}$ of characteristic zero. All rings considered are Noetherian and there is no harm in assuming them to be essentially of finite type over $\mathbb{K}$.

2.1 Generalized matroid determinants

In this section, we give a brief introduction to discriminants from the point of view of projective duality, an approach pioneered in [GKZ94]. We then recall the construction of the principal A-determinant E_A as well as its matroidal analogue, namely, the principal matroid determinant E_L introduced in [MT26] and its generalizations.

Definition 2.1. Let $X \subseteq \mathbb{P}^n$ be an irreducible projective variety. Its *dual variety* is the irreducible variety

$$X^\vee := \overline{\{\, H \in (\mathbb{P}^n)^* \mid \exists x \in X_{\mathrm{reg}} \text{ such that } H \supseteq T_x X \,\}}.$$

If $\operatorname{codim}(X^\vee) = 1$, then there exists an irreducible polynomial Δ_X, unique up to scaling, such that $X^\vee = \mathrm{V}(\Delta_X)$ called the *X-discriminant*. If $\operatorname{codim} X^\vee > 1$ then we set $\Delta_X := 1$.

X-discriminants will be the building blocks of the principal determinants. Another important construction is that of the X-resultant. Recall that for any $0 \leq k \leq n$, $\mathbb{G}(k, \mathbb{P}^n)$ parametrizes projective linear spaces of dimension k in $\mathbb{P}^n$. Suppose $\dim X = d$. Then, generically, a linear space $[\Lambda] \in \mathbb{G}(n-d-1, \mathbb{P}^n)$, of codimension $d+1$, will not intersect X. The *Chow variety* of X is formed by those linear spaces which do:

$$\mathcal{C}(X) := \{\, [\Lambda] \in \mathbb{G}(n-d-1, \mathbb{P}^n) \mid \Lambda \cap X \neq \emptyset \,\}.$$

The Chow variety is an irreducible hypersurface in $\mathbb{G}(n-d-1,\mathbb{P}^n)$ [GKZ94, Proposition 2.2]. It can be defined by a single polynomial $\mathrm{Ch}(X)$ in the Plücker variables, uniquely up to the Plücker relations. This polynomial of the same degree as X is the *Chow form* of X. There is another natural system of coordinates on the Grassmannian. Namely, consider the rational map

$$r\colon \mathbb{K}^{(d+1)\times(n+1)} \dashrightarrow \mathbb{G}(n-d-1,\mathbb{P}^n), \qquad U \mapsto [\mathbb{P}(\operatorname{Ker} U)],$$

which is defined on the dense open subset of full-rank matrices, the (non-compact) Stiefel manifold. The entries of U are called the *dual Stiefel coordinates* on the Grassmannian; the map r essentially amounts to taking all $(d+1)$-minors of U. Writing the Chow form $\mathrm{Ch}(X)$ in these coordinates yields the X-resultant.

Definition 2.2. The *X-resultant* is $\mathrm{Res}(X) \coloneqq r^*\,\mathrm{Ch}(X) \in \mathbb{K}[u_{ij} \mid 0 \le i \le d, 0 \le j \le n]$.

The case where X is a toric variety has been extensively studied in [GKZ94]. We give here a short overview of some of the main constructions and results in this setting.

Definition 2.3. Let $A \in \mathbb{Z}^{(d+1)\times(n+1)}$ be of rank $d+1$, such that $(1,\dots,1) \in \mathrm{Row}(A)$. Let $a_0,\dots,a_n \in \mathbb{K}^{d+1}$ be the columns of A. The projective toric variety X_A associated to A is the closure of the image of the map

$$(\mathbb{K}^\times)^{d+1} \to \mathbb{P}^n, \qquad x \mapsto [x^{a_0} : \cdots : x^{a_n}]$$

where $x^{a_i} \coloneqq x_0^{a_{i,0}} \cdots x_d^{a_{i,d}}$ are the Laurent monomials with exponent vectors a_i.

The X_A-discriminant and the X_A-resultant are known as the *A-discriminant* and *A-resultant*, respectively. The *principal A-determinant* is then defined as a specialization of the A-resultant.

Definition 2.4. For $z \in \mathbb{K}^{n+1}$ consider the matrix $A_z \coloneqq A \cdot \mathrm{diag}(z) \in \mathbb{K}^{(d+1)\times(n+1)}$ and let $i\colon \mathbb{K}^{n+1} \to \mathbb{K}^{(d+1)\times(n+1)}$ be the map linear map $z \mapsto A_z$. The *principal A-determinant* is

$$E_A \coloneqq i^*\,\mathrm{Res}(X_A) = i^* r^*\,\mathrm{Ch}(X_A) \in \mathbb{K}[z_0,\dots,z_n].$$

For $\mathbb{K} = \mathbb{C}$, Esterov showed that the zero locus of E_A consists of all the points $z \in (\mathbb{P}^n)^*$ such that the topological Euler characteristic of the hyperplane section $|\chi(X_A \cap \mathbb{P}((\mathbb{C}^\times)^{n+1}) \cap H_z)|$ is less than its generic value [Est13, Theorem 1.8], see also [FMT24, Section 3]. Here, H_z is the hyperplane $\mathrm{V}(z_0x_0 + \cdots + z_nx_n) \subseteq \mathbb{P}^n$.

The prime factorization of E_A can be described in terms of the lattice polytope $P_A \coloneqq \mathrm{Conv}(a_0,\dots,a_n) \subset \mathbb{R}^{d+1}$ associated to X_A. Recall that a toric variety admits a stratification by torus orbits. These strata are identified with faces of P_A via the orbit-cone correspondence, and they give rise to the factors of E_A. Let $\mathrm{Vol}(-)$ denotes the (normalized) lattice volume.

Theorem 2.5 ([GKZ94, Theorem 10.1.2])**.** *The principal A-determinant has degree* $\deg(E_A) = (d+1)\mathrm{Vol}(P_A)$. *It factors as*

$$E_A = \prod_{Q \le P_A} \Delta(X_{A|_Q})^{m_Q}$$

where Q ranges over the faces of P_A, and $m_Q = [P_A : Q]\mathrm{Vol}(Q)$.

For other varieties with a similar stratification, one may hope to construct a polynomial with similar properties to the principal A-determinant. This idea was pursued by Matsubara-Heo and Telen in the construction of the *principal matroid determinant* in [MT26], which replaces toric varieties with reciprocal linear spaces.

Definition 2.6. Let $k \in \mathbb{Z} \setminus \{0\}$, and let $L \subset \mathbb{P}^n$ be a linear space not contained in any coordinate hyperplane. The *k-th power map* is

$$\phi_k \colon L \dashrightarrow \mathbb{P}^n, \qquad x \mapsto x^k := [x_0^k : \cdots : x_n^k].$$

Define $L^k := \overline{\phi_k(L)}$: we say that L^{-1} is a *reciprocal linear space*.

Notice that ϕ_k is well-defined everywhere for $k \geq 1$, but only rational for negative values of k. We will say that the L^{-k}-discriminant $\Delta(L^{-k})$ is the *k-th generalized matroid discriminant* of L. For $k = 1$, this is the matroid discriminant defined in [MT26].

From now on, we will fix $L \subset \mathbb{P}^n$ to be a linear space of dimension d, not contained in any coordinate hyperplane, and a full-rank matrix $A \in \mathbb{K}^{(d+1)\times(n+1)}$ such that $L = \mathbb{P}(\operatorname{Row} A)$. Let $a_0, \ldots, a_n$ be the columns of A. Recall that the matroid M associated to L is the matroid on the ground set $E := \{0, \ldots, n\}$ whose independent sets index linearly independent columns of A. The matroid M does not depend on the choice of the representative matrix A. We will freely use standard terminology for matroids and refer to [Oxl11] for the precise definitions. We write $\mathcal{F}(M)$ for the set of all flats of M and $c(M)$ for the number of connected components of M. Furthermore $\mu(M)$, $\mu^+(M) := |\mu(M)|$ and $\beta(M)$ denote the signed and unsigned Möbius invariant and the beta invariant of M, respectively.

In [PS06, Proposition 5], it was shown that reciprocal linear spaces admit a stratification into pieces isomorphic to (open parts of) smaller reciprocal linear spaces, which can be described in terms of the matroid M associated to L. Namely, for $F \subset E$, let

$$T_F := \{\, x \in \mathbb{P}^n \mid x_i = 0 \text{ iff } i \notin F \,\}, \qquad \Lambda_F := \overline{T_F} = \{\, x \in \mathbb{P}^n \mid x_i = 0 \text{ if } i \notin F \,\}.$$

Then, Proudfoot and Speyer showed

$$L^{-1} = \bigsqcup_{F \in \mathcal{F}(L)} L^{-1} \cap T_F \qquad \text{and} \qquad L^{-1} \cap \Lambda_F = \overline{L^{-1} \cap T_F} \cong L_F^{-1},$$

where L_F is the projection of L onto the coordinate subspace $\mathbb{K}^F \subseteq \mathbb{K}^E$ indexed by F. In fact, as shown in [MT26, Proposition 3.10], this decomposition holds for arbitrary negative powers of linear spaces, $k \geq 1$:

$$L^{-k} = \bigsqcup_{F \in \mathcal{F}(L)} L^{-k} \cap T_F \qquad \text{and} \qquad L^{-k} \cap \Lambda_F = \overline{L^{-k} \cap T_F} \cong L_F^{-k}.$$

By slight abuse of notation, we will thus write $L_F^{-k} = \overline{L^{-k} \cap T_F}$. We can mirror the whole construction of the principal A-determinant in this setting. Recall that A is a matrix defining

Definition 2.7. Consider again the map $i\colon \mathbb{K}^{n+1} \to \mathbb{K}^{(d+1)\times(n+1)}$, $z \mapsto A \cdot \operatorname{diag}(z)$. For $k \geq 2$, the *k-th generalized matroid determinant* is

$$E_{L,-k} \coloneqq i^* \operatorname{Res}(L^{-k}) = i^* r^* \operatorname{Ch}(L^{-k}).$$

Remark 2.8. The behavior of $i^* r^* \operatorname{Ch}(L^k)$ for $k = -1$ or $k \geq 1$ is very different to that of the principal matroid determinant $E_{L,-k}$, so we decided *not* to include those cases in the definition of the generalized matroid determinant, although the definition still makes sense. We include some results and open questions for this range of k in Section 4.

Since $E_{L,-k}(z)$ is clearly homogeneous in z, it makes sense to consider its zero locus in projective space. For $k = 2$, we recover the principal matroid determinant E_L introduced in [MT26]. For $\mathbb{K} = \mathbb{C}$, Matsubara-Heo and Telen show that the zero locus of E_L consists of all the points $z \in (\mathbb{P}^n)^*$ such that the signed topological Euler characteristic of the hyperplane section $(L^{-1} \cap T_E) \cap H_z$ is less than its generic value [MT26, Theorem 6.1].

As shown in this article, the prime factorization of $E_{L,-k}$ is similar to the one of E_A described in Theorem 2.5. The role of P_A and its faces is now played by the matroid $M = \mathrm{M}(L)$ and its flats. The following result describes this factorization up to determining the multiplicities; it is a preliminary result to our main Theorem A.

Theorem 2.9. *Let $k \geq 2$. The k-th generalized matroid determinant has degree $\deg(E_{L,-k}) = (d+1)k^{d-c(M)+1}\mu^+(M)$. It factors into irreducible factors as*

$$E_{L,-k} = \prod_{F \text{ conn. flat of } M} \Delta(L_F^{-k+1})^{m_F}, \qquad m_F \geq 1.$$

For $k = 1$, this is the statement of [MT26, Theorem 1.2(1)+(2)] (minus the restriction to connected flats). Our proof is very similar, however, it is complicated by the difficulties that ϕ_k is not an isomorphism on the torus and by the fact that L^{-k} can have singularities in the torus T_E.

Proof. The degree computation follows by exactly the same argument as [MT26, Theorem 3.5], simply replacing second powers by k-th powers. As mentioned, the factorization proof follows closely the proof of [MT26, Theorem 3.11]. Consider the incidence variety

$$Y \coloneqq \left\{ (x,z) \in \mathbb{P}^n \times (\mathbb{P}^n)^* \;\middle|\; A_z \cdot x = 0 \text{ and } x \in L^{-k} \right\}.$$

The proof of [MT26, Lemma 3.7] does not require smoothness on the torus and therefore by exactly the same argument we obtain that the zero locus $E_{L,-k}$ is the image of Y under the second projection $\pi_2\colon \mathbb{P}^n \times (\mathbb{P}^n)^* \to (\mathbb{P}^n)^*$. On the other hand Y can be stratified according to the decomposition of L^{-k} into flat strata:

$$Y = \bigsqcup_{F \in \mathcal{F}(M)} \underbrace{Y \cap (T_F \times (\mathbb{P}^n)^*)}_{=: Y_F}. \tag{1}$$

We proceed in three steps to complete the proof:

(a) For any flat F we have $(L_F^{-k+1})^\vee \subseteq \overline{\pi_2(Y_F)}$.

(b) If F is a connected flat of M, then equality holds in (a) and $\overline{\pi_2(Y_F)}$ is a hypersurface.

(c) If F is a disconnected flat and F' is a component of F, then $\overline{\pi_2(Y_F)} \subseteq \overline{\pi_2(Y'_F)}$.

Once these three claims are proven the result follows: Since by the above $\pi_2(Y)$ is the vanishing locus of $E_{L,-k}$ while by (1) we also have

$$\mathrm{V}(E_{L,-k}) = \pi_2(Y) = \bigcup_{F\in\mathcal{F}(M)} \overline{\pi_2(Y_F)} \overset{\text{(c)}}{=} \bigcup_{F \text{ conn. flat of } M} \overline{\pi_2(Y_F)} \overset{\text{(b)}}{=} \bigcup_{F \text{ conn. flat of } M} (L_F^{-k+1})^\vee.$$

This yields all the irreducible factors of $E_{L,-k}$, as the dual hypersurfaces $(L_F^{-k+1})^\vee = \mathrm{V}(\Delta(L_F^{-k+1}))$ on the right hand side are irreducible. It remains to prove (a)-(c).

(a) Write $\ell_0, \dots, \ell_n$ for the linear forms on $\mathbb{K}^{d+1}$ whose coefficients in the standard basis are the columns of the matrix A, that is, $\ell(x) = x \cdot A$. Then L_F^{-k+1} is rationally parametrized by the map $\ell_F^{-k+1} \colon \mathbb{P}^d \dashrightarrow \mathbb{P}^n$ whose i-th coordinate is ℓ_i^{-k+1} if $i \in F$ and 0 otherwise. Similarly, L_F^{-k} is rationally parametrized by ℓ_F^{-k}. Assume $t \in \mathbb{P}^d$ is such that ℓ_F^{-k+1} is defined at t and $\ell_F^{-k+1}(t)$ is a smooth point of L_F^{-k+1}. Notice that in particular ϕ_{-k+1} is unramified at t, so we can use the parametrization ℓ_F^{-k+1} to compute the tangent space to L_F^{-k+1} at $\ell_F^{-k+1}(t)$. The proof of [MT26, Lemma 3.6] then shows that the pair $(\ell_F^{-k}(t), z)$ belongs to Y_F if and only if the hyperplane defined by z contains the tangent space to L_F^{-k+1} at $\ell_F^{-k+1}(t)$. This implies that $(L_F^{-k+1})^\vee \subseteq \overline{\pi_2(Y_F)}$.

(b) Notice that $A_z \cdot x = A_x \cdot z$ and hence the fiber in Y over a fixed point $x \in L^{-k}$ is a linear space Λ_x. The dimension of Λ_x is $n - \mathrm{rk}(A_x)$ and hence it only depends on which coordinates of x vanish, since a non-zero rescaling of columns of A does not alter the rank. More precisely we have $\mathrm{rk}(A_x) = \mathrm{rk}_M(F)$ whenever $x \in T_F$. In particular, Y_F the total space is a vector bundle over $L^{-k} \cap T_F$ and therefore irreducible of dimension

$$\dim Y_F = \dim L_F^{-k+1} + \dim \Lambda_x = \mathrm{rk}_M(F) - 1 + n - \mathrm{rk}_M(F) = n - 1.$$

In particular $\overline{\pi_2(Y_F)}$ is an irreducible variety of dimension at most $n-1$. On the other hand, assuming F is connected, we have that $(L_F^{-k+1})^\vee$ is a hypersurface by [BKW26, Theorem 4.8]. Hence equality must hold in (a) in that case.

(c) Suppose F is disconnected and write $F = F' \sqcup G$ where F' is a connected flat of M. By an invertible change of coordinates in $\mathrm{GL}_{d+1}(\mathbb{K})$ we can assume that the columns of A whose index belongs to F are of block shape; this does not change Y. This means A is of the form displayed below where the first columns belong to F', the second set of columns belongs to G and the last set is in F^c.

$$A = \begin{bmatrix} A_1 & 0 & A_3 \\ 0 & A_2 & A_4 \end{bmatrix} \quad \begin{matrix} \underbrace{}_{F'} & \underbrace{}_{G} & \underbrace{}_{F^c} \end{matrix}$$

Now suppose $z \in \pi_2(Y_F)$, then there exists $x \in T_F \cap L^{-k+1}$ such that $A_z \cdot x = 0$. But then setting the entries x_i with $i \in G$ to zero we obtain a point $x' \in L^{-k+1} \cap T_{F'}$ which still satisfies $A_z \cdot x' = 0$ due to the block shape of A. Hence $z \in \pi_2(Y_{F'})$ which finishes the proof. □

The description of $E_{L,-k}$ in Theorem 2.9 is incomplete without a closed formula for the exponents m_F. Our main result Theorem A provides such a formula; Section 3 is devoted to the proof of this theorem.

2.2 Local multiplicity

We recall some properties of local multiplicity needed for the proof of Theorem A. For reference, see for example [ZS60, Chapter VII.10] or [SH06, Chapter 11]. All rings appearing in this section are assumed to be Noetherian.

Let $(R, \mathfrak{m}_R)$ be a Noetherian local ring and I an $\mathfrak{m}_R$-primary ideal. Under this assumption, the quotients I^i/I^{i+1} are R-modules of finite length. The Hilbert-Samuel function of (R, I) is the Hilbert function of the associated graded ring $\mathrm{gr}_I(R) \coloneqq \bigoplus_{i\geq 0} I^i/I^{i+1}$

$$\chi_R^I(i) \coloneqq \mathrm{hf}_{\mathrm{gr}_I(R)}(i) \coloneqq \mathrm{len}_R(I^i/I^{i+1}).$$

This function agrees with the Hilbert polynomial $\mathrm{HP}_{I,R}(t) \in \mathbb{Q}[t]$ for $i \gg 0$. If $\dim R = n$, then $\deg \mathrm{HP}_{I,R} = n - 1$.

Definition 2.10 (Local multiplicity)**.** The multiplicity of the $\mathfrak{m}_R$-primary ideal $I \subset R$, denoted by $e(I, R) \in \mathbb{N}_{>0}$, is the normalized leading coefficient of the Hilbert polynomial

$$\mathrm{HP}_{I,R}(t) = \frac{e(I,R)}{(n-1)!}t^{n-1} + O(t^{n-2}).$$

Equivalently, writing the Hilbert series $\sum_{i\geq 0} \chi_R^I(i)t^i$ as $\frac{Q(t)}{(1-t)^n}$, then $e(I, R) = Q(1)$.

For any Noetherian ring R, $I \subseteq R$ a $\mathfrak{p}$-primary ideal, we set $e(I, R) \coloneqq e(IR_\mathfrak{p}, R_\mathfrak{p})$.

If X is a variety, or more generally an algebraic $\mathbb{K}$-scheme, and $Z \subseteq X$ a closed irreducible subset, then the *local multiplicity* of X along Z is $\mathrm{mult}_Z(X) \coloneqq e(\mathfrak{m}_{X,Z}, \mathcal{O}_{X,Z})$ where $\mathfrak{m}_{X,Z}$ denotes the maximal ideal of the local ring $\mathcal{O}_{X,Z}$ associated to Z.

The local multiplicity satisfies the following properties, which we will use later on.

Proposition 2.11. (i) *Let $J \subset I \subset R$ be $\mathfrak{m}_R$-primary ideals, then $e(I, R) \leq e(J, R)$.*

(ii) *Let $f : R \to R'$ be an étale morphism of local rings with the same residue field, and let $I \subset R$ be an $\mathfrak{m}_R$-primary ideal. Denote $I' \coloneqq f(I)R' \subset R'$. Then $e(I, R) = e(I', R')$.*

(iii) *Let R_1, R_2 be $\mathbb{K}$-algebras, $\mathfrak{m}_i \subseteq R_i$ maximal ideals with $R_i/\mathfrak{m}_i = \mathbb{K}$, and let $I_i \subseteq R_i$ be $\mathfrak{m}_i$-primary. Let $I = I_1 \otimes R_2 + R_1 \otimes I_2 \subseteq R \coloneqq R_1 \otimes R_2$, then I is $(\mathfrak{m}_1 \otimes R_2 + R_1 \otimes \mathfrak{m}_2)$-primary and $e(I, R) = e(I_1, R_1)e(I_2, R_2)$.*

Proof. (i) Assume $e(I,R) > e(J,R)$, then $\chi^I_R(t) = \chi^J_R(t) + \frac{e(I,R)-e(J,R)}{(n-1)!}t^{n-1} + O(t^{n-2})$, contradicting the estimate

$$\sum_{j=0}^{i} \chi^I_R(j) = \operatorname{len}_R(R/I^i) \leq \operatorname{len}_R(R/J^i) = \sum_{j=0}^{i} \chi^J_R(j).$$

(ii) Since R and R' have the same residue fields, f being étale implies that f induces an isomorphism of the completions of R and R', see [Liu02, Proposition 4.3.26]. The statement follows from the fact that Hilbert-Samuel multiplicity is preserved under completion [ZS60, Chapter VII, Lemma 8.1].

(iii) With the given assumptions and notation, we have an isomorphism of associated graded algebras

$$\operatorname{gr}_I(R) = \operatorname{gr}_{I_1 \otimes R_2 + R_1 \otimes I_2}(R \otimes R') = \operatorname{gr}_{I_1}(R) \otimes \operatorname{gr}_{I_2}(R').$$

Let $\mathfrak{m} = \mathfrak{m}_1 \otimes R_2 + R_1 \otimes \mathfrak{m}_2$, then the Hilbert series of I, I_1, I_2 satisfy

$$\operatorname{HS}_{I_{\mathfrak{m}}, R_{\mathfrak{m}}} = \operatorname{HS}_{(I_1)_{\mathfrak{m}_1}, (R_1)_{\mathfrak{m}_1}} \cdot \operatorname{HS}_{(I_2)_{\mathfrak{m}_2}, (R_2)_{\mathfrak{m}_2}}.$$

From the description $e(I,R) = ((1-t)^{\dim R_{\mathfrak{m}}} \operatorname{HS}_{I_{\mathfrak{m}}, R_{\mathfrak{m}}}(t))(1)$, we see that the multiplicities multiply. □

The associated graded ring of the local ring $\mathcal{O}_{X,x}$ of a point of a scheme is the coordinate ring of the tangent cone $\operatorname{TC}_x X = \operatorname{Spec} \operatorname{gr}_{\mathfrak{m}_{X,x}}(\mathcal{O}_{X,x})$. In particular, the Hilbert function and -polynomial of $\mathcal{O}_{X,x}$ are that of the projectivized tangent cone $\mathbb{P}(\operatorname{TC}_x X) \hookrightarrow \mathbb{P}(\operatorname{T}_x X)$, so by definition $\operatorname{mult}_x X = \deg \mathbb{P}(\operatorname{TC}_x X)$.

Example 2.12. Let $X = \widehat{L^{-1}} \subseteq \mathbb{A}^E$ and let $x \in X$ be in the stratum $X \cap T_F$ corresponding to the flat F (that is, $x_i \neq 0$ if and only if $i \in F$). In [SSV13, Theorem 24], the authors show that the tangent cone $\operatorname{TC}_0(X - x) \subseteq \operatorname{TC}_0 \mathbb{A}^E = \mathbb{A}^E$ has the explicit description

$$\operatorname{TC}_0(X - x) = \widehat{L|_F} \times \widehat{(L/F)^{-1}} \subseteq \mathbb{A}^F \times \mathbb{A}^{E \setminus F}.$$

In particular, since $\widehat{L|_F}$ is a linear space, $\mathbb{P}(\operatorname{TC}_x X)$ is a cone over $(L/F)^{-1}$, hence

$$\operatorname{mult}_x X = \deg \mathbb{P}(\operatorname{TC}_x X) = \deg (L/F)^{-1} = \mu^+(M/F). \qquad \diamondsuit$$

Geometrically the local multiplicity enjoys the following properties, which we will use throughout the article:

Proposition 2.13. *Let X be an algebraic scheme, $Z \subseteq X$ an irreducible closed subset.*

(i) *If X is reduced and equidim. at $x \in X$, then $\operatorname{mult}_x X = 1$ if and only if $x \in X_{\mathrm{reg}}$.*

(ii) *If X is equidimensional, then the map $Y \mapsto \operatorname{mult}_Y(X)$ is upper semi-continuous on X (including its non-closed points). In particular,*

$$\operatorname{mult}_Z(X) = \min_{p \in Z} \operatorname{mult}_p X = \operatorname{mult}_x X \text{ for } x \text{ in a dense open } U \subseteq Z.$$

(iii) *If Y is another scheme, $x \in X$ and $y \in Y$, then* $\operatorname{mult}_{(x,y)} X \times Y = \operatorname{mult}_x X \cdot \operatorname{mult}_y Y$.

(iv) *If $X \subseteq \mathbb{P}^n$ is (quasi-)projective, then* $\operatorname{mult}_Z X = \operatorname{mult}_{\widehat{Z}} \widehat{X}$.

(v) *Let $W \subseteq X$ be a subscheme locally around Z defined by $f \in \mathcal{O}_{X,Z}$, that is, $\mathcal{I}(W)\mathcal{O}_{X,Z} = f\mathcal{O}_{X,Y}$. Assume that $\mathcal{O}_{X,Z}$ is regular, then*

$$\operatorname{mult}_Z(W) = \operatorname{ord}_{\mathfrak{m}_{X,Z}}(f) \coloneqq \max\{\, k \in \mathbb{N} \mid f \in \mathfrak{m}_{X,Z}^k \,\}.$$

(vi) *If $X \subseteq \mathbb{P}^n$ is irreducible of dimension d and $x \in X$, then* $\operatorname{mult}_x X = i(x, X \cdot \Lambda, \mathbb{P}^n)$, *where $[\Lambda] \in \mathbb{G}(n-d, \mathbb{P}^n)$ is a general linear space containing x. Here i denotes the intersection number of the cycles X and Λ at p.*

Proof. (i) This is evident from the previous tangent cone description. The more general algebraic statement is due to Nagata [Nag62, Theorem 40.6]: A Noetherian local ring is regular if and only if $e(R) = 1$ and A is quasi-unmixed ($\dim \widehat{R}/\mathfrak{p} = \dim \widehat{R}$ for all associated primes of $0_{\widehat{R}}$). The latter condition is always satisfied for local rings of varieties, so $\operatorname{mult}_x(X) = 1$ if and only if $\mathcal{O}_{X,x}$ is regular.

(ii) The semi-continuity of local multiplicity is attributed to Dade [Dad60]. A precise statement can be found in [Vil14, Theorem 6.12], where the result is discussed in the generality of excellent schemes.

(iii) This is a geometric reformulation of Proposition 2.11(iii).

(iv) Locally, $\widehat{X}$ is isomorphic to an open set of $X \times \mathbb{A}^1$. More precisely, $\widehat{X}$ is the open subset of the total space of $\mathcal{O}_X(-1)$ given by the complement of the zero section. Hence we can (locally) apply (iii), using that $\mathbb{A}^1$ is regular.

(v) This is [SH06, Example 11.2.8].

(vi) This is [Ful98, Example 12.4.5]. □

Example 2.14. Let $W = \mathrm{V}(f) \subseteq \mathbb{A}^n$ be a hypersurface, and consider the irredundant factorization $f = f_1^{e_1} \cdots f_m^{e_m}$ with f_i irreducible. Let $Y = \mathrm{V}(f_i)$, then $f\mathcal{O}_{\mathbb{A}^n,Y} = f_i^{e_i}\mathcal{O}_{\mathbb{A}^n,Y} = \mathfrak{m}_{\mathbb{A}^n,Y}^{e_i}$, so $\operatorname{mult}_Y(W) = e_i$. In this way, the local multiplicity recovers the standard notion of algebraic multiplicity of a factor in a polynomial. ♢

We end this section with a theorem which will later be useful.

Theorem 2.15 ([ZS60, Section VIII.10, Theorem 24 & Corollary 1])**.** *Let R be a local ring with maximal ideal $\mathfrak{m}_R$ and consider a finite extension of rings $R \hookrightarrow S$. Let $\mathfrak{n}_1, \ldots, \mathfrak{n}_r$ be the maximal ideals of the semi-local S. Suppose no nonzero element of R is a zero divisor in S and for all $i = 1, \ldots, r$, the local ring $S_{\mathfrak{n}_i}$ has the same dimension as R.*

Let I be an ideal of R which is $\mathfrak{m}_R$-primary and let J_i be the $\mathfrak{n}_i$-primary component of SI, then

$$[S : R] \cdot e(I, R) = \sum_{i=1}^{r} [S/\mathfrak{n}_i : R/\mathfrak{m}_R] \cdot e(J_i, S).$$

If $R/\mathfrak{m}_R$ is algebraically closed field, then the residue field extensions are trivial and the right hand side simplifies accordingly. If R is the local ring of a smooth curve, then this recovers the classical degree formula for (ramified) covers of algebraic curves.

3 Proof of the main theorem

In this section we give the proof of Theorem A. We start by illustrating this Theorem in two examples.

Remark 3.1. By [BKW26, Theorem 4.8], for $k \geq 2$, $\Delta(L_F^{-k+1})$ is dual-defective if and only if F is not a connected flat, in which case it does not contribute to the factorization of $E_{L,-k}$. Therefore, the product in the factorization of $E_{L,-k}$ in Theorem A may be taken either over connected flats only or over all flats, where the disconnected flats simply contribute a factor of 1. In particular, the component $\Delta(L^{-1})$ appears non-trivially in the factorization if and only if $\mathrm{M}(L)$ is connected. It corresponds to the flat $F = E$, and so it appears with multiplicity

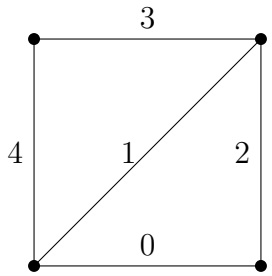
$$m_E = k^{\mathrm{rk}(M/E)}\mu^+(M/E) = 1$$

as M/E is the empty matroid.

Example 3.2 (A graphical example). Let $L = \mathbb{P}(\mathrm{Row}\, A) \subset \mathbb{P}^4$, where

$$A = \begin{bmatrix} 1 & 0 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & -1 \end{bmatrix}.$$

This matrix has rank $d+1$, its matroid $M = \mathrm{M}(L)$ is the graphic matroid associated with the following graph.



By Remark 3.1, since M is connected, we already know that $m_E = 1$. Furthermore, the connected flats of M are the singletons, the whole ground set, and the rank 2 flats of size 3 ($\{0,1,2\}$ and $\{1,3,4\}$). We compute the multiplicities of each of these flats. For concreteness, we let $k = 2$.

connected flat F	$\mathrm{rk}(F)$	$\mathrm{rk}(M/F)$	$\mu^+(M/F)$	m_F	$2^{\mathrm{rk}\,F-1}\beta(M\|_F)$
$\{0\},\{2\},\{3\},\{4\}$	1	2	2	8	1
$\{1\}$	1	2	1	4	1
$\{0,1,2\},\{1,3,4\}$	2	1	1	2	2
$\{0,1,2,3,4\}$	3	0	1	1	4

The principal matroid determinant $E_{L,-2}$ factors as

$$E_{L,-2}(z) = \Delta(L^{-1})\,\Delta(L^{-1}_{\{0,1,2\}})^2\,\Delta(L^{-1}_{\{1,3,4\}})^2\,z_0^8\,z_1^4\,z_2^8\,z_3^8\,z_4^8 \in \mathbb{K}[z].$$

This matches degree-wise: $E_{L,-2}$ has degree

$$(d+1)2^d\mu^+(M) = 48 = 4 + 2\cdot 2 + 2\cdot 2 + 8 + 4 + 8 + 8 + 8. \qquad \diamondsuit$$

Example 3.3 (Uniform linear spaces). Let $L \subset \mathbb{P}^n$ be a general linear space of dimension d. The matroid associated to L is the uniform matroid of rank $d+1$ on $n+1$ elements $U_{d+1,n+1}$. The only connected flats of $U_{d+1,n+1}$ are the entire ground set E and the singletons, therefore

$$E_{L,-k} = \Delta(L^{-k+1})^{m_E}\prod_{i=0}^{n}\Delta(L^{-k+1}_{\{i\}})^{m_{\{i\}}}.$$

Now, $L^{-k+1}_{\{i\}} = L_{\{i\}}$ is the i-th coordinate point. Its dual variety is the set of hyperplanes containing that point, so $\Delta(L_{\{i\}}) = z_i \in \mathbb{K}[z]$. By Theorem A,

$$m_{\{i\}} = k^{\mathrm{rk}(M/\{i\})-c(M)+1}\mu^+(M/\{i\}) = k^d\mu^+(U_{d+1,n}) = k^d\binom{n-1}{d-1}.$$

As above by Remark 3.1 we have $m_E = 1$. We obtain in this way the factorization of $E_{L,-k}$:

$$E_{L,-k} = (z_0z_1\cdots z_{n-1}z_n)^{k^d\binom{n-1}{d-1}}\Delta(L^{-k+1}).$$

The reader may check that using the following formulas (see [BKW26, Example 4.2] and Theorem 2.9),

$$\deg\Delta(L^{-k+1}) = k^d\binom{n-1}{d}, \qquad \deg E_{L,-k} = (d+1)k^d\binom{n}{d},$$

the following equality holds:

$$\deg E_{L,-k} = \deg\Delta(L^{-k+1}) + (n+1)k^d\binom{n-1}{d}. \qquad \diamondsuit$$

For the rest of this section we will fix $L \subseteq \mathbb{P}^n$ a linear space of dimension d not contained in any coordinate hyperplane, and an integer $k \in 2$.

3.1 A lower bound

In this section we show that the local multiplicities of L^{-k} along its flat strata are lower bounds on the exponents m_F appearing in the factorization of $E_{L,-k}$. To do so, we pass through the Chow incidence correspondence. Since the methods in this section work in greater generality than just for powers of reciprocal linear spaces, we fix an irreducible projective variety $X \subseteq \mathbb{P}^n$ of dimension d and an irreducible closed subset $Y \subseteq X$.

We start by introducing some notation. Let $\mathcal{I}$ be the *Chow incidence correspondence*

$$\mathcal{I} \coloneqq \{\, (x, [\Lambda]) \mid x \in X \cap \Lambda \,\} \subseteq X \times \mathbb{G}(n-d-1, \mathbb{P}^n).$$

The projection $\pi_1 \colon \mathcal{I} \twoheadrightarrow X$ is surjective, while the image of $\pi_2 \colon \mathcal{I} \to \mathbb{G}(n-d-1, \mathbb{P}^n)$ is the Chow variety $\mathcal{C}(X)$. Let $\mathcal{C}_X(Y)$ be the image of $\mathcal{I} \cap (Y \times \mathbb{G}(n-d-1, \mathbb{P}^n))$ under π_2, set-theoretically

$$\mathcal{C}_X(Y) = \{\, [\Lambda] \mid Y \cap \Lambda \neq \emptyset \,\} \subseteq \mathbb{G}(n-d-1, \mathbb{P}^n).$$

Note that $\mathcal{C}_X(Y) \ncong \mathcal{C}(Y)$ (they don't have the same dimension), though the latter is a Grassmannian bundle over the former.

Proposition 3.4. *With the above notations,* $\operatorname{mult}_Y X = \operatorname{mult}_{\mathcal{C}_X(Y)} \mathcal{C}(X)$.

We note that the later considerations for $X = L^{-k}$ only require the (easier) inequality $\operatorname{mult}_Y X \leq \operatorname{mult}_{\mathcal{C}_X(Y)} \mathcal{C}(X)$. The proof of this inequality passes through the incidence $\mathcal{I}$, while the reverse inequality relies on the fact that X and $\mathcal{C}(X)$ have the same degree (the latter in the Plücker embedding), plus an application of Bertini's and Bézout's theorem.

Proof. We have $\operatorname{mult}_Y X = \min_{p \in Y} \operatorname{mult}_p X$ (Proposition 2.13(ii)) and similarly

$$\operatorname{mult}_{\mathcal{C}_X(Y)} \mathcal{C}(X) = \min_{\Lambda \in \mathcal{C}_X(Y)} \operatorname{mult}_\Lambda \mathcal{C}(X) = \min_{\substack{p \in Y \\ \Lambda \in \mathcal{C}_X(p)}} \operatorname{mult}_\Lambda \mathcal{C}(X) = \min_{p \in Y} \operatorname{mult}_{\mathcal{C}_X(p)} \mathcal{C}(X),$$

so we may without loss of generality assume that Y is a point $p \in X$. Let $\Lambda \in \mathcal{C}_X(p)$ be general in the sense of Proposition 2.13(ii), i.e. $\operatorname{mult}_\Lambda \mathcal{C}(X) = \operatorname{mult}_{\mathcal{C}_X(p)} \mathcal{C}(X)$. We first claim that

$$\operatorname{mult}_p(X) \overset{\text{(a)}}{=} \operatorname{mult}_{(p,\Lambda)}(\mathcal{I}) \overset{\text{(b)}}{\leq} \operatorname{mult}_\Lambda(\mathcal{C}(X)).$$

(a) Over any point $x \in X$, the fiber $\pi_1^{-1}(\{x\})$ consists of all $n-d-1$-dimensional subspaces of $\mathbb{P}^n$ that contain x. Writing $\mathbb{P}^n = \mathbb{P}(\mathbb{K}^{n+1})$ and identifying x with a line in $\mathbb{K}^{n+1}$, the fiber is hence $\mathbb{G}(n-d-2, \mathbb{P}(\mathbb{K}^{n+1}/x))$. In other words, the first projection π_1 turns $\mathcal{I}$ into the relative Grassmannian bundle $\mathbb{G}(n-d-2, \mathcal{Q}_X)$ over X, where $\mathcal{Q}_X$ is the universal quotient bundle of $\mathbb{P}^n$ restricted to X. In particular π_1 trivializes Zariski-locally on X, i.e. we can find an open affine neighborhood $U \subset X$ of p such that

$$\pi_1^{-1}(U) \cong U \times \mathbb{G}(n-d-2, \mathbb{P}^{n-1})$$

Let (p, Λ') be the image of (p, Λ) under the above identification. By Proposition 2.13(iii),

$$\begin{aligned} \operatorname{mult}_{(p,\Lambda)} \mathcal{I} &= \operatorname{mult}_{(p,\Lambda')}(U \times \mathbb{G}(n-d-2, \mathbb{P}^{n-1})) \\ &= (\operatorname{mult}_p U) \cdot (\operatorname{mult}_{\Lambda'} \mathbb{G}(n-d-2, \mathbb{P}^{n-1})) \\ &= \operatorname{mult}_p X \cdot 1, \end{aligned}$$

where we use that the Grassmannian is smooth (Proposition 2.13(i)).

(b) The dominant projection $\pi_2 \colon \mathcal{I} \twoheadrightarrow \mathcal{C}(X)$ induces the inclusion of local rings

$$R \coloneqq \mathcal{O}_{\mathcal{C}(X),\Lambda} \hookrightarrow S \coloneqq \mathcal{O}_{\mathcal{I},(p,\Lambda)}.$$

Let $\mathfrak{m}_R, \mathfrak{m}_S$ be the maximal ideals of R, S respectively. We define $J \coloneqq \mathfrak{m}_R S$ as the ideal generated by $\mathfrak{m}_R$ in S under the above inclusion. Since X is irreducible, so are $\mathcal{C}(X)$ and $\mathcal{I}$ and hence R, S are integral domains. The morphism $\pi_2^{-1}(\mathcal{C}_X(p)) \to \mathcal{C}_X(p)$ is generically injective (a general linear space from the Grassmannian containing p intersects X only in p), so $R \hookrightarrow S$ is finite generically free of rank 1 and J is $\mathfrak{m}_R$-primary. We can now apply Theorem 2.15:

$$e(\mathfrak{m}_R, R) = \underbrace{[S : R]}_{=1} e(\mathfrak{m}_R, R) = \underbrace{[S/\mathfrak{m}_S : R/\mathfrak{m}_R]}_{=[\mathbb{K}:\mathbb{K}]=1} e(J, S) = e(J, S) \geq e(\mathfrak{m}_S, S).$$

Here the last inequality follows from Proposition 2.11(i) using $J \subseteq \mathfrak{m}_S$. This concludes the proof of (b) since by definition $e(\mathfrak{m}_R, R) = \operatorname{mult}_\Lambda \mathcal{C}(X)$ and $e(\mathfrak{m}_S, S) = \operatorname{mult}_{(p,\Lambda)} \mathcal{I}$.

We have established the inequality $\operatorname{mult}_p X \leq \operatorname{mult}_{\mathcal{C}_X(p)} \mathcal{C}(X)$; it remains to show the reverse inequality. Let $M \in \mathbb{G}(n-d, \mathbb{P}^n)$ be a general linear space containing p, then by Bertini's theorem [Kol+23, Theorem 11.1.2], $M \cap X = \{p, x_1, \ldots, x_r\}$ with X and M intersecting transversally at x_i. By Bézout's theorem for projective varieties [Ful98, Proposition 8.4] we have

$$\deg X = \deg(X \cap M) = \sum_{x \in X \cap M} i(x, X \cdot M, \mathbb{P}^n) = \operatorname{mult}_p X + r, \tag{2}$$

where the last equality also uses the characterization Proposition 2.13(v). Now consider a general subspace $[L] \in \mathbb{G}(n-d-2, M)$ and the pencil of linear spaces

$$P \coloneqq \{\, [\Lambda] \in \mathbb{G}(n-d-1, \mathbb{P}^n) \mid L \subseteq \Lambda \subseteq M \,\} = \mathbb{P}(M/L) \cong \mathbb{P}^1.$$

Let $q \in M \setminus \langle L, p \rangle$ be any point, then P can be given coordinates $[s : t] \in \mathbb{P}^1$ corresponding to $[\langle L, sp+tq \rangle]$. Let f be the restriction of the Chow form $\operatorname{Ch}(X)$ to $P \hookrightarrow \mathbb{G}(n-d-1, \mathbb{P}^n)$. This is a binary form of degree $\deg f = \deg \operatorname{Ch}(X) = \deg X$ which vanishes on $[\Lambda]$ if and only if $\Lambda \cap X \neq \emptyset$. One of the roots of f is $[1:0] \mathrel{\hat{=}} [\langle L, p \rangle]$ corresponding to p, the roots $[\Lambda_i]$ of f correspond to the other intersection points $x_1, \ldots, x_r \in X \cap M$. The local multiplicity of $\mathcal{C}(X)$ at $\Lambda = \langle L, p \rangle$ is at most the multiplicity of f at $[1:0]$ by Lemma 3.5 below, so we have

$$\operatorname{mult}_{[\Lambda]} \mathcal{C}(X) \overset{3.5}{\leq} \operatorname{mult}_{[1:0]} f = \deg X - \sum_{i=1}^{r} \operatorname{mult}_{[\Lambda_i]} f \leq \deg X - r \overset{(2)}{=} \operatorname{mult}_p X.$$

Since $[\Lambda]$ is general in $\mathcal{C}_X(p)$, we conclude $\operatorname{mult}_{\mathcal{C}_X(p)} \mathcal{C}(X) \leq \operatorname{mult}_p X$. □

Lemma 3.5. *Let $\varphi \colon X' \to X$ be a morphism of smooth varieties. Let $Y \subseteq X$ be a subvariety and assume that Y and its scheme-theoretic preimage $\varphi^{-1}(Y) \subset X'$ have codimension 1. For all $p \in \varphi^{-1}(Y)$ we have*

$$\operatorname{mult}_p \varphi^{-1}(Y) \geq \operatorname{mult}_{\varphi(p)} Y.$$

Since Λ is general in $\mathcal{C}_Y(X)$, this finishes the proof.

Proof. The map φ induces a local ring map $\varphi^*\colon \mathcal{O}_{X,\varphi(p)} \to \mathcal{O}_{X',p}$. Let $f \in \mathcal{O}_{X,\varphi(p)}$ be a regular function defining Y in a neighbourhood of $\varphi(p)$, then $\varphi^*(f) \in \mathcal{O}_{X',p}$ is a regular function defining $\varphi^{-1}(Y)$ in a neighbourhood of p. Since X and X' are smooth, $\mathcal{O}_{X,\varphi(p)}$ and $\mathcal{O}_{X',p}$ are regular so we are in a situation to apply Proposition 2.13(v). For all $e \geq 0$ we have $\varphi^*(\mathfrak{m}^e_{X,\varphi(p)}) \subseteq \mathfrak{m}^e_{X',p}$, so if $f \in \mathfrak{m}^e_{X,\varphi(p)}$ then also $\varphi^*(f) \in \mathfrak{m}^e_{X',p}$. In other words, $\operatorname{ord}_{\mathfrak{m}_{X',p}}(\varphi^*(f)) \geq \operatorname{ord}_{\mathfrak{m}_{X,\varphi(p)}}(f)$; applying the proposition yields

$$\operatorname{mult}_p \varphi^{-1}(Y) = \operatorname{ord}_{\mathfrak{m}_{X',p}}(\varphi^*(f)) \geq \operatorname{ord}_{\mathfrak{m}_{X,\varphi(p)}}(f) = \operatorname{mult}_{\varphi(p)} Y. \qquad \square$$

We now apply the above lemmata to our specific situation. For a flat F of $M = \mathrm{M}(L)$, recall that $T_F = \{\, [p_0 : \cdots : p_n] \in \mathbb{P}^n \mid p_i = 0 \text{ iff } i \notin F \,\} \subseteq \mathbb{P}^n$ and $\Lambda_F = \overline{T_F}$.

Corollary 3.6. *Let $k \geq 2$. For every flat F of M we have the inequality*

$$\operatorname{mult}_{L_F^{-k}}(L^{-k}) \leq m_F$$

where m_F denotes the multiplicity of the factor $\Delta(L_F^{-k+1})$ in the polynomial $E_{L,-k}$.

Proof. Denote $\mathcal{E}_{L,-k} \coloneqq \mathrm{V}(E_{L,-k})$, then we have $m_F = \operatorname{mult}_{(L_F^{-k+1})^\vee} \mathcal{E}_{L,-k}$ (Example 2.14). We first apply Proposition 3.4 to $X = L^{-k}$ and $Y = L_F^{-k}$ and obtain that

$$\operatorname{mult}_{L_F^{-k}}(L^{-k}) = \operatorname{mult}_{\mathcal{C}_{L^{-k}}(L_F^{-k})} \mathcal{C}(L^{-k}).$$

Now consider the rational map

$$\varphi \coloneqq r \circ i\colon \mathbb{P}^n \dashrightarrow \mathbb{G}(n-d-1, \mathbb{P}^n), \qquad z \mapsto [\operatorname{Ker}(A \cdot \operatorname{diag}(z))].$$

and notice that by construction, $\mathcal{E}_{L,-k}$ is the scheme-theoretic preimage of $\mathcal{C}(L^{-k})$ under this map (on its domain of definition). Pick $p \in (L_F^{-k+1})^\vee$ general as in Proposition 2.13(ii), such that $\operatorname{mult}_p \mathcal{E}_{L,-k} = \operatorname{mult}_{(L_F^{-k+1})^\vee} \mathcal{E}_{L,-k}$. Notice that, as seen in in the proof of Theorem 2.9, we have

$$\overline{\varphi((L_F^{-k+1})^\vee)} = \mathcal{C}_{L^{-k}}(L_F^{-k})$$

and therefore $\varphi(p) \in \mathcal{C}_{L^{-k}}(L_F^{-k})$. Since both $\mathcal{C}(L^{-k})$ and $\mathcal{E}_{L,-k}$ are hypersurfaces, we can apply Lemma 3.5 to $X = \mathbb{G}(n-d-1, \mathbb{P}^n)$, $X' = \mathbb{P}^n$, $\varphi = r \circ i$. We to obtain

$$\operatorname{mult}_p(\mathcal{E}_{L,-k}) \geq \operatorname{mult}_{\varphi(p)}(\mathcal{C}(L^{-k})) \geq \operatorname{mult}_{\mathcal{C}_{L^{-k}}(L_F^{-k})} \mathcal{C}(L^{-k}),$$

where the second inequality is again upper semi-continuity of local multiplicity (Proposition 2.13(ii)). This finishes the proof. $\square$

3.2 Local multiplicities of L^{-k}

Next, we compute the multiplicity $\operatorname{mult}_{L_F^{-k}}(L^{-k})$. An important ingredient is the degree of the coordinatewise k-th power map after restricting to L^{-1}, which we briefly recall here. Remember that $c(M)$ denotes the number of connected components of M.

Lemma 3.7. *Consider the coordinatewise k-th power map*

$$h_k : L^{-1} \to L^{-k}, \qquad x \mapsto x^k.$$

This morphism is finite of $\deg(h_k) = k^{c(M)-1}$. *In particular, the corresponding map of affine cones* $\widehat{L^{-1}} \to \widehat{L^{-k}}$ *is finite to one with degree* $k^{c(M)}$

Proof. The map is finite since it is already finite on $\mathbb{P}^n$. Since the rational Cremona map which inverts coordinates is one-to-one, the degree of h_k equals the degree of $\widetilde{h}_k : L \to L^k, x \mapsto x^k$. This degree can be easily computed using [DGK20, Theorem 2.6]; for more details we refer to the proof of [BKW26, Lemma 3.2], where the argument is carried out for the case $k = 2$. The additional factor of k in the affine case comes from the fact that each affine representative of a point in a fixed fiber can be rescaled by the k different k-th roots of unity which does not change the projective point. □

Proposition 3.8. *For every flat F of M we have the equality*

$$\operatorname{mult}_{L_F^{-k}}(L^{-k}) = k^{\operatorname{rk}(M/F)-c(M)+c(M|_F)}\mu^+(M/F).$$

Proof. Without loss of generality we may assume $F = \{0, \dots, s\}$ for some $s < n$. By Proposition 2.13(iv) we have $\operatorname{mult}_{L_F^{-k}}(L^{-k}) = \operatorname{mult}_{\widehat{L_F^{-k}}}(\widehat{L^{-k}})$ and hence we pass to affine cones. Fix $q \in \widehat{L_F^{-k}}$ such that both

- $\operatorname{mult}_{\widehat{L_F^{-k}}}(\widehat{L^{-k}}) = \operatorname{mult}_q(\widehat{L^{-k}})$, see Proposition 2.13(ii), and
- in coordinates we have $q = (q_0, \dots, q_s, 0, \dots, 0)$ for $q_0, \dots, q_s \in \mathbb{K}^\times$, i.e. $q \in \widehat{T_F}$.

The surjective k-th power map $\widehat{L^{-1}} \to \widehat{L^{-k}}$ induces the following inclusion of affine coordinate rings:

$$R' \coloneqq \mathbb{K}[y_0, \dots, y_n]/I(\widehat{L^{-k}}) \hookrightarrow S' \coloneqq \mathbb{K}[x_0, \dots, x_n]/I(\widehat{L^{-1}}), \qquad y_i \mapsto x_i^k.$$

Since the k-th power map is finite, S' is a finitely generated R'-module under this map.

The maximal ideal corresponding to q is $\widetilde{\mathfrak{m}}_q = \langle y_1 - q_1, \dots y_s - q_s, y_{s+1}, \dots, y_n\rangle$. Localizing at this maximal ideals we obtain the inclusion

$$R \coloneqq \left(\mathbb{K}[y_1, \dots, y_n]/I(\widehat{L^{-k}})\right)_{\widetilde{\mathfrak{m}}_q} \hookrightarrow S \coloneqq \left(\mathbb{K}[x_1, \cdots x_n]/I(\widehat{L^{-1}})\right) \otimes_{R'} R$$

We set $\mathfrak{m}_q \coloneqq \widetilde{\mathfrak{m}}_q R$. Under the inclusion of R into S we define $J \coloneqq \mathfrak{m}_q S$. S is a semi-local ring with maximal ideals $\mathfrak{m}_p = \langle x_0 - p_0, \dots, x_s - p_s, x_{s+1}, \dots, x_n\rangle_S$ for any point $p \in \widehat{L^{-1}}$ with $p^k = q$. Write J_p for the $\mathfrak{m}_p$-primary component of J, then we have

$$\begin{aligned} J_p S_{\mathfrak{m}_p} &= \langle x_0^k - q_0, \dots, x_s^k - q_s, x_{s+1}^k, \dots, x_n^k\rangle_{S_{\mathfrak{m}_p}} \\ &= \langle (x_0 - p_0)f_0, \dots, (x_s - p_s)f_s, x_{s+1}^k, \dots, x_n^k\rangle_{S_{\mathfrak{m}_p}} \\ &= \langle x_0 - p_0, \dots, x_s - p_s, x_{s+1}^k, \dots, x_n^k\rangle_{S_{\mathfrak{m}_p}}, \end{aligned}$$

where $f_i \coloneqq \sum_{j=0}^{k} x_i^j p_i^{k-j}$. In the last step we used that the polynomials f_i are in the complement of the maximal ideal $\mathfrak{m}_p$ and hence are units in the local ring $S_{\mathfrak{m}_p}$.

By choice of q we have $\operatorname{mult}_{L_F^{-k}}(L^{-k}) = \operatorname{mult}_q(\widehat{L^{-k}}) = e(\mathfrak{m}_q, R)$ and by Theorem 2.15 we get

$$[S : R] \cdot e(\mathfrak{m}_q, R) = \sum_{\substack{p \in L^{-1} \\ p^k = q}} \underbrace{[S/\mathfrak{m}_p : R/\mathfrak{m}_q]}_{=[\mathbb{K}:\mathbb{K}]=1} \cdot e(J_p, S).$$

Here $[S : R]$ equals the degree of the finite k-th power map on affine cones, which is $k^{c(M)}$ by Lemma 3.7. Therefore it suffices to show $e(J_p, S) = k^{\operatorname{rk}(M/F)}\mu^+(M/F)$ for every $p \in \widehat{L^{-1}}$ with $p^k = q$. We defer this lengthy computation to Lemma 3.9. Indeed applying again Lemma 3.7, there are precisely $k^{c(M|_F)}$ preimages $p \in \widehat{L^{-1}}$ of a general point $q \in \widehat{L^{-k} \cap T_F}$ under the k-th power map. Hence, given the result of Lemma 3.9, we get

$$k^{c(M)} \operatorname{mult}_{L_F^{-k}}(L^{-k}) = \sum_{\substack{p \in \widehat{L^{-1}} \\ p^k = q}} k^{\operatorname{rk}(M/F)}\mu^+(M/F) = k^{c(M|_F)+\operatorname{rk}(M/F)}\mu^+(M/F),$$

which finishes the proof after dividing by $k^{c(M)}$ on both sides. □

Lemma 3.9. *Using the notations of the proof of Proposition 3.8, fix $p \in \widehat{L^{-1}}$ such that $p^k = q$. We have that*

$$e(J_p, S) = k^{\operatorname{rk}(M/F)}\mu^+(M/F).$$

Proof. By definition $e(J_p, S) = e(J_p S_{\mathfrak{m}_p}, S_{\mathfrak{m}_p})$. By [EPW16, Theorem 3.3] there exists an open set $W_F^\circ \subset \widehat{L^{-1}}$ containing p and a map

$$\phi : W_F^\circ \to \widehat{(L_F)^{-1}} \times \widehat{(L/F)^{-1}}$$

which sends p to $((p_0, \dots, p_s), 0)$ and is étale at p. Here $L/F \coloneqq L \cap \bigcap_{i \in F}\{x_i = 0\} \subseteq \mathbb{P}(\mathbb{K}^{F^c})$ denotes the intersection of L with the coordinate hyperplanes corresponding to F. It induces a map of local rings $\phi^\# \colon M_{\widetilde{\mathfrak{m}}_p} \to S_{\mathfrak{m}_p}$ where

$$M \coloneqq \underbrace{(\mathbb{K}[z_0, \dots, z_s]/I(\widehat{L_F^{-1}}))_{\langle z_0 - p_0, \dots, z_s - p_s\rangle}}_{=: M_1} \otimes_{\mathbb{K}} \underbrace{(\mathbb{K}[z_{s+1}, \dots, z_n]/I(\widehat{(L/F)^{-1}}))_{\langle z_{s+1}, \dots, z_n\rangle}}_{=: M_2}$$

$$\widetilde{\mathfrak{m}}_p \coloneqq \langle z_0 - p_0, \dots, z_s - p_s, z_{s+1}, \dots, z_n\rangle_M.$$

Consider the ideal $J_p' \coloneqq \langle z_0 - p_0, \dots, z_s - p_s, z_{s+1}^k, \dots, z_n^k\rangle_M \subseteq M$, the ideal generated by $\phi^\#(J_p' M_{\widetilde{\mathfrak{m}}_p})$ in $S_{\mathfrak{m}_p}$ is then

$$\begin{aligned}
\phi^\#(J_p' M_{\widetilde{\mathfrak{m}}_p}) S_{\mathfrak{m}_p} &= \langle \phi^\#(z_0 - p_0), \dots, \phi^\#(z_s - p_s), \phi^\#(z_{s+1}^k), \dots, \phi^\#(z_n^k)\rangle_{S_{\mathfrak{m}_p}} \\
&= \langle \phi^\#(z_0) - p_0, \dots, \phi^\#(z_s) - p_s, \phi^\#(z_{s+1})^k, \dots, \phi^\#(z_n)^k\rangle_{S_{\mathfrak{m}_p}} \\
&= J_p.
\end{aligned}$$

The last equality here uses the construction of ϕ ([EPW16, Theorem 3.3]):

$$\phi^{\#}(z_i) = \begin{cases} x_i & 0 \leq i \leq s \\ \dfrac{x_i}{1 - b_i x_i} & s < i \leq n, \end{cases}$$

where $b_i \in \mathbb{K}[\widehat{L^{-1} \cap T_F}]$. Notice that $1 - b_i x_i \in S_{\mathfrak{m}_p} \setminus \mathfrak{m}_p$, so it is a unit in $S_{\mathfrak{m}_p}$.

Since the map $\phi^{\#}$ is étale, by Proposition 2.11(ii),

$$e(J_p, S_{\mathfrak{m}_p}) = e(\phi^{\#}(J'_p M_{\widetilde{\mathfrak{m}}_p}) S_{\mathfrak{m}_p}, S_{\mathfrak{m}_p}) = e(J'_p M_{\widetilde{\mathfrak{m}}_p}, M_{\widetilde{\mathfrak{m}}_p}) = e(J'_p, M).$$

Now since J'_p splits as $J'_p = J_1 \otimes_{\mathbb{K}} M_2 + M_1 \otimes_{\mathbb{K}} J_2$, where

$$J_1 = \langle z_0 - p_0, \ldots, z_s - p_s \rangle \subseteq M_1, \qquad J_2 = \langle z_{s+1}^k, \ldots, z_n^k \rangle \subseteq M_2,$$

by Proposition 2.11(iii), we obtain $e(J'_p, M) = e(J_1, M_1) e(J_2, M_2)$. By Example 2.12, since $(p_0, \ldots, p_s) \in \widehat{T_F} \cap \widehat{(L_F)^{-1}}$,

$$e(J_1, M_1) = \mu^+((M|_F)/F) = \mu^+(\emptyset) = 1.$$

Alternatively this follows from Proposition 2.13(i) since the point $(p_0, \ldots, p_s)$ has non-zero coordinates and is therefore a smooth point of $(L_F)^{-1}$. It remains to determine $e(J_2, M_2)$.

Let $J_3 \coloneqq \langle z_{s+1}, \ldots, z_n \rangle \subseteq M_2$. Notice that for any $l \geq 0$, $(J_3^k)^{l+1}$ is generated by all monomials of degree $k(l+1)$, while $(J_3^k)^l J_2$ is generated by all monomials of degree $k(l+1)$ such that at least one variable appears with exponent greater or equal to k. If $k(l+1) > (k-1)(n-s)$, these two generating sets coincide. Since additionally $J_3^k \supseteq J_2$, J_2 is a *reduction* of J_3^k in the sense of Rees [SH06, Definition 1.2.1]. It follows that $e(J_2, M_2) = e(J_3^k, M_2)$ [SH06, Proposition 11.2.1].

We wish to relate $e(J_3^k, M_2)$ to $e(J_3, M_2)$ which is simply the degree of $(L/F)^{-1}$. To do so, note that for any $i = 0, \ldots, k-2$ and any $l > 0$ we have

$$(J_3^{kl+i} / J_3^{kl+k}) / (J_3^{kl+i+1} / J_3^{kl+k}) = J_3^{kl+i} / J_3^{kl+i+1}$$

as M_2-modules and hence in particular as $\mathbb{K}$-vector spaces. We therefore get

$$\dim_{\mathbb{K}}(J_3^{kl+i} / J_3^{kl+k}) = \dim_{\mathbb{K}}(J_3^{kl+i} / J_3^{kl+i+1}) + \dim_{\mathbb{K}}(J_3^{kl+i+1} / J_3^{kl+k}).$$

Combining these equations for all $i = 0, \ldots, k-2$ we deduce

$$\dim_{\mathbb{K}}((J_3^k)^l / (J_3^k)^{l+1}) = \dim_{\mathbb{K}}(J_3^{kl} / J_3^{kl+k}) = \sum_{i=0}^{k-1} \dim_{\mathbb{K}}(J_3^{kl+i} / J_3^{kl+i+1}).$$

For $l \gg 0$ this implies the following equality of Hilbert polynomials:

$$\mathrm{HP}_{J_3^k, M_2}(t) = \sum_{i=0}^{k-1} \mathrm{HP}_{J_3, M_2}(kt + i) \in \mathbb{Z}[t].$$

All Hilbert polynomials appearing in this equation have degree $\dim M_2 - 1$, hence by comparing leading terms and multiplying by $(\dim(M_2) - 1)!$ we obtain

$$e(J_3^k, M_2) = \sum_{i=0}^{k-1} k^{\dim M_2 - 1} e(J_3, M_2) = k^{\dim M_2} e(J_3, M_2).$$

Here, $\dim M_2 = \dim \widehat{(L/F)^{-1}} = \mathrm{rk}(M/F)$ and $e(J_3, M_2)$ is the local multiplicity of the affine cone over $(L/F)^{-1}$. This degree is $\mu^+(M/F)$ by Example 2.12, and hence we conclude that

$$e(J_2, M_2) = e(J_3^k, M_2) = k^{\mathrm{rk}(M/F)} \mu^+(M/F). \qquad \square$$

3.3 Matching the lower bound

In this section we finish the proof of Theorem A by showing that the lower bound we derived in the previous sections is actually an equality. The main ingredient will be the following lemma.

Lemma 3.10. *Let M be any matroid, then*

$$\mathrm{rk}(M) \cdot \mu^+(M) = \sum_F \beta(M|_F) \cdot \mu^+(M/F),$$

where the summation run over all subsets, equivalently all connected flats of M.

For the proof we recall the fundamentals of Möbius inversion for matroids [Whi87, Chapter 7]. Let f, g be functions that take as input a matroid and return a complex number. Their convolution product is defined as

$$(f * g)(M) \coloneqq \sum_{F \text{ flat of } M} f(M|_F) g(M/F).$$

This product is associative, *not* commutative, and satisfies *Möbius inversion*:

$$(\mu * 1)(M) = \delta_{\emptyset, M}, \qquad (1 * \mu)(M) = \delta_{0, \mathrm{rk}(M)}.$$

Proof of Lemma 3.10. Since $\beta(M|_F)\mu^+(M/F) = 0$ unless F is a connected flat, we may restrict our attention to the summation over all flats of M. The equation in Lemma 3.10 then asserts that $\mathrm{rk} \cdot \mu^+ = \beta * \mu^+$. Define the "signed β-invariant" as

$$\beta^{\pm}(M) \coloneqq (-1)^{\mathrm{rk}\, M} \beta(M) = -\chi'_M(1) = \sum_{F \text{ flat of } M} \mu(M|_F) \,\mathrm{rk}(M|_F).$$

Replacing $\mu, \beta^{\pm}$ by μ^+, β, our desired identity then reads $\mathrm{rk} \cdot \mu = \beta^{\pm} * \mu$. The last sum in the defining equation of $\beta^{\pm}$ shows that $\beta^{\pm} = (\mathrm{rk} \cdot \mu) * 1$. Applying $* \mu$ and using Möbius inversion yields the desired formula

$$\beta^{\pm} * \mu = (\mathrm{rk} \cdot \mu) * 1 * \mu = (\mathrm{rk} \cdot \mu) * \delta_{0, \mathrm{rk}(-)} = \mathrm{rk} \cdot \mu. \qquad \square$$

Proof of Theorem A. By definition of the m_F in Theorem 2.9,

$$\deg(E_{L,-k}) = \sum_{F \text{ conn. flat of } M} m_F \cdot \deg(L_F^{-k+1})^\vee.$$

Combining Corollary 3.6 and Proposition 3.8, for any flat F of M,

$$m_F \geq \operatorname{mult}_{L_F^{-k}}(L^{-k}) = k^{\operatorname{rk}(M/F)-c(M)+1}\mu^+(M/F).$$

We also know from [BKW26, Theorem 4.8] that $\deg(L_F^{-k+1})^\vee = k^{\operatorname{rk}(M|_F)-1}\beta(M|_F)$ whenever F is a connected flat, hence

$$\begin{aligned} m_F \cdot \deg(L_F^{-k+1})^\vee &\geq k^{\operatorname{rk}(M/F)-c(M)+1}\mu^+(M/F) \cdot k^{\operatorname{rk}(M|_F)-1}\beta(M|_F) \\ &= k^{\operatorname{rk}(M)-1-c(M)+1}\mu^+(M/F)\beta(M|_F). \end{aligned} \tag{3}$$

Therefore, using our notation $d = \operatorname{rk} M - 1$,

$$\begin{aligned} \deg(E_{L,-k}) &\geq \sum_{F \text{ conn. flat of } M} k^{d-c(M)+1}\mu^+(M/F)\beta(M|_F) \\ &= k^{d-c(M)+1} \sum_{F \text{ flat of } M} \mu^+(M/F)\beta(M|_F) \\ &= k^{d-c(M)+1} \operatorname{rk}(M)\mu^+(M) \end{aligned}$$

where the last equality uses Lemma 3.10. But we already know from Theorem 2.9 that $\deg(E_{L,-k}) = k^{d-c(M)+1}\operatorname{rk}(M)\mu^+(M)$. Therefore, the above inequality is in fact an equality, and thus the inequalities in equations (3) for *all flats* F must be equalities too. In particular,

$$m_F = \operatorname{mult}_{L_F^{-k}}(L^{-k}) = k^{\operatorname{rk}(M/F)-c(M)+1}\mu^+(M/F)$$

for all connected flats F of M. □

4 Other powers of linear spaces

We briefly give some results on the generalised matroid determinant $E_{L,k}$ when $k > -2$. We start with the case $k = -1$ where, as before, $E_{L,-1} \coloneqq i^* r^* \mathrm{Ch}(L^{-1})$ for the maps i, r from Definition 2.7.

Proposition 4.1. *The polynomial $E_{L,-1}$ has degree $\deg(E_{L,-1}) = (d+1)\mu^+(M)$ and factorizes as*

$$E_{L,-1}(z) = \prod_{F \text{ conn. flat of } M} \Big(\sum_{i \in F} z_i\Big)^{m_{F,-1}} \in \mathbb{K}[z]$$

for some integers $m_{F,-1} \geq 1$.

Proof. The degree formula for $E_{L,-1}$ follows from a similar argument as [MT26, Theorem 3.5]:

$$\deg(E_{L,-1}) = \deg(i)\deg(r)\deg(\mathrm{Ch}(L^{-1})) = (d+1)\deg(L^{-1}) = (d+1)\mu^+(M).$$

For the second claim, fix a flat $F \in \mathcal{F}(M)$, and define

$$Y_F \coloneqq \{\, (x,z) \in \mathbb{P}^n \times (\mathbb{P}^n)^* \mid A_z \cdot x = 0 \text{ and } x \in L^{-1} \cap T_F \,\}.$$

Since $x_i = 0$ for $i \notin F$, the condition $A_z \cdot x = 0$ becomes

$$\sum_{j \in F} a_{ij} x_j z_j = 0, \qquad i = 0, \dots, d,$$

or, equivalently, $(x_i z_i)_{i \in F}$ is in $L_F^\perp \coloneqq \{\, y \in \mathbb{P}(\mathbb{K}^F) \mid \forall w \in L_F : \sum_{i \in F} y_i w_i = 0 \,\}$. Notice that for a fixed $z \in (\mathbb{P}^n)^*$ with the property that not all $z_i = 0$ for $i \in F$, there exists $x \in L^{-1} \cap T_F$ with the above property if and only if $(z_i)_{i \in F} \in (L_F \cap T_F) \star L_F^\perp$. This shows that $\overline{\pi_2(Y_F)} = J(L_F \star L_F^\perp, \mathbb{P}(\mathbb{K}^{F^c}))$ where J denotes the join of two projective varieties (see [BKW26, Section 2.2] for more context).

Now, the stratification of L^{-1} by flats induces the stratification

$$Y \coloneqq \bigsqcup_{F \in \mathcal{F}(M)} Y_F \subseteq \mathbb{P}^n \times (\mathbb{P}^n)^*$$

whose image under the projection $\pi_2 \colon \mathbb{P}^n \times (\mathbb{P}^n)^* \to (\mathbb{P}^n)^*$ is the vanishing locus of $E_{L,-1}$ by the same reasoning as in [MT26, Lemma 3.7]. The above argument shows that

$$\mathrm{V}(E_{L,-1}) = \pi_2\Big(\bigsqcup_{F \in \mathcal{F}(M)} Y_F \Big) = \bigcup_{F \in \mathcal{F}(M)} \overline{\pi_2(Y_F)} = \bigcup_{F \in \mathcal{F}(M)} J(L_F \star L_F^\perp, \mathbb{P}(\mathbb{K}^{F^c})).$$

Now, by [MT26, Proposition 4.6] and [BKW26, Theorem 2.11], $\dim(L_F \star L_F^\perp) = |F| - 2$ if and only if F is a connected flat, and in that case, it is the hyperplane given by $\sum_{i \in F} z_i = 0$. For disconnected flats F we have that $\overline{\pi_2(Y_F)}$ is contained in $\overline{\pi_2(Y_{F'})}$ for any connected component F' of F, see also the proof Theorem 2.9. This completes the argument. □

Remark 4.2. Define $L^0 \coloneqq \{\mathbf{1}\}$ where $\mathbf{1}$ is the point with all coordinates equal to one. Then,

$$E_{L,-1} = \prod_{F \text{ conn. flat of } M} \Delta(L_F^0)^{m_F}$$

for some integers $m_F \geq 0$, matching the general shape for the factorization of the generalized matroid determinant in Theorem 2.9.

The proof of our main theorem does not extend to the case $E_{L,-1}$. In fact all arguments in Sections 3.1 and 3.2 can be adapted also to this case, but the obtained inequality $m_{F,-1} \geq \mathrm{mult}_{L_F^{-1}} L^{-1} = \mu^+(M/F)$ is in general strict. However, there exists a determinantal representation for the Chow form of L^{-1} due to Kummer & Vinzant [KV19, Corollary 3.9, Remark 3.3]. A close inspection of the determinantal formula reveals that $E_{L,-1}$, equivalently the $m_{F,-1}$, depend only on the (labeled) matroid $\mathrm{M}(L)$ and *not* on the realization L. Using the formula, we verified the following conjecture for uniform matroids and computationally for small non-uniform matroids.

Conjecture 4.3. The multiplicities in Proposition 4.1 satisfy

$$m_{F,-1} = \mu^+(M/F)\beta(M|_F)$$

for every connected flat F of M.

Finally, we consider the case of positive powers L^k, $k \geq 1$. The same degree computation as in the case -1 shows the following.

Lemma 4.4. *Let $k \in \mathbb{Z}_{\geq 1}$ and fix a linear subspace $L \subseteq \mathbb{P}^n$ be of dimension d such that its matroid has no loops. Set $E_{L,k} \coloneqq i^* r^* \operatorname{Ch}(L^k)$ as in Definition 2.7. Then*

$$\deg(E_{L,k}) = (d+1)k^{d-c(M)+1}.$$

However, the factorization behavior seems to be quite different from the case of negative powers: Based on computational experiments, it seems that, assuming M is connected, $E_{L,k}$ is an *irreducible* polynomial equal to the defining equation of the dual hypersurface $(L^{k+1})^\vee$. A similar method as has been used in the proof of Theorem 2.9 can show that the zero locus of $E_{L,k}$ indeed still contains $(L^{k+1})^\vee$. We note also that, in this case, analogous results from [BKW26] (most notably the dual degree of L^k), which we have used throughout this article, have not yet been established, compare [BKW26, Remark 4.13].

Conjecture 4.5. Let $k \geq 1$ and let $L \subseteq \mathbb{P}^d$ be of dimension d and such that its matroid M is connected, then $E_{L,k} = \Delta(L^{k+1})$. In particular $\deg(L^{k+1})^\vee = (d+1)k^d$.

Example 4.6. For $k = 1$, one can deduce Conjecture 4.5 from elementary considerations. A direct calculation shows that

$$\operatorname{Res}(L)(u) = \det(UA^\mathsf{T}), \qquad E_{L,1}(z) = \det(A \operatorname{diag}(z) A^\mathsf{T}) = \sum_{S \in \binom{[n+1]}{d+1}} \det(A_S)^2 z^S,$$

where the last formula is the Cauchy–Binet formula. This expansion shows that S is in the support of $E_{L,1}$ if and only if S is a basis of the matroid $M = \mathrm{M}(L)$. If M is connected, then such a polynomial must be irreducible, see [Cho+04, Proposition 4.6]. As $\Delta(L^2) \mid E_{L,1}$, these two polynomials must be equal up to scaling. ♢

Acknowledgements

C.B. thanks the Max Planck Institute for Mathematics in the Sciences for its hospitality and the Fondation de l'ENS for its financial support. L.K. was partially supported by the project "Symmetry in Large Algebraic Structures", funded by the SNSF grant 10000940. J.W. was supported by the SPP 2458 "Combinatorial Synergies", funded by the DFG grant 539677510.

References

[BKW26] Clara Briand, Leonie Kayser, and Julian Weigert. *Polar Degrees of Matroids*. 2026. arXiv: 2606.26281 [math.CO] (cit. on pp. 3, 7, 11, 12, 16, 20–22).

[Cho+04] Young-Bin Choe et al. “Homogeneous multivariate polynomials with the half-plane property”. In: *Advances in Applied Mathematics* 32.1-2 (Jan. 2004), pp. 88–187. ISSN: 0196-8858. DOI: 10.1016/s0196-8858(03)00078-2 (cit. on p. 22).

[Dad60] Everett C. Dade. *Multiplicity and monoidal transformations*. Princeton University, 1960. URL: https://www.proquest.com/docview/301924379 (cit. on p. 10).

[DGK20] Papri Dey, Paul Görlach, and Nidhi Kaihnsa. “Coordinate-wise powers of algebraic varieties”. In: *Beiträge zur Algebra und Geometrie / Contributions to Algebra and Geometry* 61.3 (Jan. 2020), pp. 473–505. ISSN: 2191-0383. DOI: 10.1007/s13366-019-00481-8 (cit. on p. 16).

[EPW16] Ben Elias, Nicholas Proudfoot, and Max Wakefield. “The Kazhdan–Lusztig polynomial of a matroid”. In: *Advances in Mathematics* 299 (2016), pp. 36–70. ISSN: 0001-8708. DOI: 10.1016/j.aim.2016.05.005 (cit. on pp. 3, 17, 18).

[Est13] Alexander Esterov. “The discriminant of a system of equations”. In: *Advances in Mathematics* 245 (2013), pp. 534–572. ISSN: 0001-8708. DOI: 10.1016/j.aim.2013.06.021 (cit. on pp. 2, 4).

[FMT24] Claudia Fevola, Sebastian Mizera, and Simon Telen. “Principal Landau determinants”. In: *Computer Physics Communications* 303 (Oct. 2024), p. 109278. ISSN: 0010-4655. DOI: 10.1016/j.cpc.2024.109278 (cit. on p. 4).

[Ful98] William Fulton. *Intersection Theory*. Springer New York, 1998. ISBN: 9781461217008. DOI: 10.1007/978-1-4612-1700-8 (cit. on pp. 10, 14).

[GKZ94] Israel M. Gelfand, Mikhail M. Kapranov, and Andrei V. Zelevinsky. *Discriminants, Resultants, and Multidimensional Determinants*. Birkhäuser Boston, 1994. ISBN: 9780817647711. DOI: 10.1007/978-0-8176-4771-1 (cit. on pp. 1–4).

[Kol+23] János Kollár et al. *What Determines an Algebraic Variety?* Princeton University Press, Aug. 2023. ISBN: 9780691246833. DOI: 10.1515/9780691246833 (cit. on p. 14).

[KV19] Mario Kummer and Cynthia Vinzant. “The Chow Form of a Reciprocal Linear Space”. In: *Michigan Mathematical Journal* 68.4 (2019), pp. 831–858. DOI: 10.1307/mmj/1571731287 (cit. on p. 21).

[Liu02] Qing Liu. *Algebraic Geometry and Arithmetic Curves*. Vol. 6. Oxford Graduate Texts in Mathematics. Oxford: Oxford University Press, 2002. ISBN: 9780198502845 (cit. on p. 9).

[MT26] Saiei-Jaeyeong Matsubara-Heo and Simon Telen. *Principal Matroid Determinants*. 2026. DOI: 10.48550/ARXIV.2604.24667 (cit. on pp. 2, 3, 5–7, 20, 21).

[Nag62] Masayoshi Nagata. *Local rings*. Vol. No. 13. Interscience Tracts in Pure and Applied Mathematics. Interscience Publishers (a division of John Wiley & Sons, Inc.), New York-London, 1962, pp. xiii+234 (cit. on p. 10).

[Oxl11] James Oxley. *Matroid Theory*. 2nd ed. Oxford University Press, 2011. ISBN: 9780199603398 (cit. on p. 5).

[PS06] Nicholas Proudfoot and David Speyer. "A broken circuit ring." eng. In: *Beiträge zur Algebra und Geometrie* 47.1 (2006), pp. 161–166. URL: http://eudml.org/doc/126347 (cit. on pp. 2, 5).

[SH06] Irena Swanson and Craig Huneke. *London mathematical society lecture note series: Integral closure of ideals, rings, and modules series number 336*. Cambridge, England: Cambridge University Press, Oct. 2006. URL: https://www.math.purdue.edu/~iswanso/book/index.html (cit. on pp. 8, 10, 18).

[SSV13] Raman Sanyal, Bernd Sturmfels, and Cynthia Vinzant. "The entropic discriminant". In: *Advances in Mathematics* 244 (Sept. 2013), pp. 678–707. ISSN: 0001-8708. DOI: 10.1016/j.aim.2013.05.019 (cit. on p. 9).

[Vil14] Orlando E. Villamayor U. "Equimultiplicity, algebraic elimination, and blowing-up". In: *Advances in Mathematics* 262 (Sept. 2014), pp. 313–369. ISSN: 0001-8708. DOI: 10.1016/j.aim.2014.04.013 (cit. on p. 10).

[Whi87] Neil White, ed. *Combinatorial Geometries*. Cambridge University Press, Sept. 1987. ISBN: 9781107325715. DOI: 10.1017/cbo9781107325715 (cit. on p. 19).

[ZS60] Oscar Zariski and Pierre Samuel. *Commutative Algebra: Volume II*. Vol. 29. Graduate Texts in Mathematics. New York-Heidelberg-Berlin: Springer-Verlag, 1960. ISBN: 978-3-662-27753-9. DOI: 10.1007/978-3-662-29244-0 (cit. on pp. 8–10).

Addresses of Clara Briand
École Normale Supérieure, France
E-mail address: clara.briand@ens.psl.eu

Addresses of Leonie Kayser
Max Planck Institute for Mathematics in the Sciences, Leipzig
and University of Bern, Switzerland
E-mail address: leonie.kayser@unibe.ch

Addresses of Julian Weigert
Max Planck Institute for Mathematics in the Sciences, Leipzig
and Georg-August-Universität Göttingen, Germany
E-mail address: julian.weigert@mis.mpg.de